Smart Dumbbell

Preliminary Specification v0.1

ECE 198 | Group 57 | Exported 18 October 2026

1 Purpose and scope

Purpose: Build a custom hollow smart dumbbell that helps users follow a prescribed bicep-curl routine, automatically counts repetitions, continuously measures heart rate during exercise, and transmits session data to a computer.

Intended users: CCCARE graduates and participants who can independently perform prescribed dumbbell exercises at home. Initial engineering tests involve team members or student volunteers; these tests do not establish suitability for all older adults.

Item	Version 1 definition
Supported exercise	Bicep curl; one instrumented dumbbell at a time.
Motion recognition	An on-device machine-learning model classifies curl, idle, and other.
Required features	Motion recognition, repetition counting, heart-rate measurement during movement, display feedback, wireless transmission, and computer logging.
Physical design	Custom removable hollow enclosure with securely mounted electronics and power supply.
Budget	Maximum CAD 100 for the team, including taxes and shipping.
Device access	The core system must not require the user to own a smartwatch. A computer is used for session configuration, review, and logging.
Excluded features	Grip-force or muscle-strength measurement, recovery prediction, full-body posture assessment, and automatic changes to exercise prescriptions.

Status and interpretation

All numerical performance requirements are **proposed acceptance targets**, not achieved results or instructor requirements. MUST identifies a required Version 1 feature. Failed requirements remain incomplete unless the team explicitly approves a scope change.

This specification describes the smart dumbbell concept. The project pitch must still introduce a second concept separately. Use simulated records or anonymous test IDs in the classroom prototype.

2 Functional requirements

All requirements on this page have MUST priority.

ID	Requirement
FR-01	Allow users to set a target repetition count on the computer and synchronize it with the dumbbell. Demonstration default: 10 repetitions.
FR-02	Provide start, pause/resume, and end controls. Repetitions must not increase while paused.
FR-03	Continuously acquire three-axis motion data with timing information for training and testing.
FR-04	Run motion-model inference on the microcontroller. The computer performs training and logging, not motion classification during normal operation.
FR-05	Increment the count only when a complete movement cycle is detected within motion classified as a curl. Count each cycle once.
FR-06	Avoid counting stationary periods, picking up or putting down the dumbbell, and unrelated movements as valid curls.
FR-07	Acquire optical signals and update heart-rate estimates during continuous exercise. Measurement only during stationary rest periods does not satisfy this requirement.
FR-08	Display Insufficient signal and record an invalid value when signal quality is inadequate, contact is lost, or data becomes stale. Do not present an old reading as current.
FR-09	Display current/target repetitions, session state, heart rate and validity, and wireless connection state.
FR-10	Provide a clear visual indication when the target is reached. Continue recording additional repetitions if the user continues exercising.
FR-11	Transmit motion and heart-rate information over Wi-Fi. Local recognition, counting, and display must continue during connection loss.
FR-12	Display live information on the computer and save exportable CSV records and a session summary.
FR-13	Schedule sensing, inference, heart-rate processing, display updates, and communication so heart-rate processing does not block motion acquisition for extended periods.

One complete repetition: Begin at the calibrated lower position, raise the dumbbell to the defined demonstration upper position, and return to the lower position. Thresholds are established through training and calibration data. A time window classified as curl is not automatically one repetition.

3 Hardware and mechanical design

Component	Qty	Source and purpose
Nano ESP32 with headers	1	Purchase; main controller, inference, and Wi-Fi.
Grove ADXL335	1	Course loan; three-axis acceleration.
GY-MAX30102	1	Course loan; evaluate heart-rate measurement during movement.
OLED display	1	Course base kit; local feedback.
Pushbuttons	3	Course base kit; start, pause/resume, and end.
Complete USB power supply	1	Reuse an existing small unit or purchase.
Enclosure and mounting materials	1 set	Custom construction with removable mounting.
R4 Minima and breadboard	1 set	Course base kit; bench development, not required inside the final dumbbell.
Reference heart-rate device	1	Preferably a borrowed chest strap capable of exporting continuous records for synchronized comparison.

Nano ESP32 supports Wi-Fi and BLE but does not include the required motion sensor or OLED. Verify exact borrowed-module versions, supply requirements, and signal voltages before connection; chip specifications alone do not establish breakout-board compatibility. [1]

Mechanical requirements

• Use two removable hollow ends connected by a handle. Provide access to USB, the power switch, and serviceable components.

• The power supply and controller may occupy opposite ends. Adjust placement using measured masses rather than assuming balance.

• Rigidly mount the accelerometer and document its axes. Components must not move freely inside the enclosure.

• Place the optical sensor where a finger naturally contacts the handle. Include light shielding and a structure that maintains contact.

• Do not add training weights in Version 1. Target a total demonstration-prototype mass of no more than 1 kg.

• Use removable mounting for borrowed equipment. Determine final CAD dimensions from measured components; validate contact and clearance before final fabrication.

4 Data flow and interfaces

Development workflow

Collect motion data → label data → train and independently test on a computer → deploy the model to Nano ESP32.

Operating workflow

Motion path: ADXL335 → motion classification → complete-cycle detection → repetition count.

Heart-rate path: MAX30102 → signal processing and quality checks → heart rate or invalid status.

Output path: Both paths feed the OLED and Wi-Fi transmission, followed by the computer interface and CSV logging.

Communication baseline

The Nano creates a local Wi-Fi access point. The computer connects to it and uses an HTTP interface to configure sessions and retrieve status. Version 1 does not depend on campus Wi-Fi or a cloud service.

Minimum fields	Meaning
session_id, sequence, elapsed_ms	Session identifier, record sequence, and elapsed time.
session_state, target_reps, rep_count	Session state, target count, and measured count.
motion_class, motion_confidence	Classification result and model confidence.
heart_rate_bpm, heart_rate_valid	Heart-rate estimate and validity; value is null when invalid.
signal_quality, connection_state	Signal quality and connection status.

Target regular status transmission and heart-rate output at once per second. Heart-rate estimates may use a rolling analysis window. Development mode additionally records raw acceleration and optical signals.

Failure handling

During disconnection, the computer marks a data gap. After reconnection, it retrieves the current cumulative count without fabricating missing heart-rate history.

A power interruption ends the session as interrupted. Version 1 does not require recovery of unsaved data.

5 Verification and acceptance

Freeze procedures before evaluation. Do not revise acceptance criteria merely to match observed results.

ID	Test	Acceptance criterion
AT-01	Classification	Macro-F1 >= 0.85 across three classes on participants/sessions excluded from training and tuning. Report the confusion matrix.
AT-02	Counting	For 20 sets of 10 complete repetitions, at least 18 sets have an error within +/-1 repetition. Include different speeds, pauses, and interspersed partial movements.
AT-03	Other movements	At least 18 of 20 predefined pickup, put-down, and unrelated-motion trials produce no false repetitions.
AT-04	Heart rate in motion	At least five participants each perform a 60-second continuous demonstration sequence. Allow <=20 seconds for stabilization, then compare synchronized readings every second.
AT-05	Heart-rate quality	For each participant: valid-output coverage >=90% and mean absolute error of valid outputs <=10 bpm. Report both; do not select only successful intervals.
AT-06	Contact loss	Mark heart rate invalid within two seconds after finger contact is lost. Do not substitute the last valid reading.
AT-07	Concurrent operation	Run recognition, heart-rate processing, display, and transmission together for 10 minutes without resets or freezes. Record acquisition losses.
AT-08	Network interruption	Disconnect the computer for 30 seconds. Local counting continues; cumulative count agrees after reconnection; the log identifies the disconnected interval.
AT-09	Usability	After one demonstration, at least four of five test users start a session within two minutes.
AT-10	Mechanics and runtime	Complete 100 lightweight demonstration repetitions without detached components, sensor displacement, or loose handle connections. Support >=30 minutes of wireless operation.

Split training, validation, and test data by **participant or complete recording session**. Do not randomly distribute overlapping windows from one recording across training and test sets.

Movement introduces optical interference, making AT-04 through AT-06 early feasibility priorities. Without a reference device, heart-rate development may continue, but accuracy cannot be accepted. A stationary test does not replace the movement test. [2]

6 Budget and delivery

Item	Allocation CAD
Nano ESP32	28
USB power supply	15
Enclosure fabrication and mounting	20
Additional connection materials	8
Taxes, shipping, and contingency	29
Total limit	100

This budget assumes course sensors, the OLED, and a reference heart-rate device can be borrowed, and a computer is available. The enclosure allocation is not a fabrication quotation. If costs exceed the limit, revise procurement or request a budget change before purchasing.

Work sequence

1. User and market research. One person extracts CCCARE requirements, compares three to four existing solutions, and documents evidence supporting the design.

2. Technical and hardware research. A second person confirms borrowed-module versions, model deployment compatibility, heart-rate acquisition methods, and prices.

3. Feasibility testing. Prioritize heart rate through the handle during movement and ADXL335 classification/counting. Retain raw data.

4. Bench integration. Run the model, heart-rate processing, display, and wireless logging concurrently on the final controller.

5. Enclosure construction. Build around validated sensor placement and measured dimensions. Collect additional training data where mounting changes affect signals.

6. System acceptance. Execute AT-01 through AT-10. Report each requirement as passed, failed, or untested.

Deliverables and change control

Deliver one prototype, firmware, training scripts and model, a computer logging application, bill of materials, enclosure design files, test data, and a results report.

All selected features remain required. If motion recognition, heart rate during exercise, or budget constraints cannot be met, propose an explicit specification change for team review. Do not describe incomplete features as implemented.

Technical references

[1] Arduino. Nano ESP32 documentation. docs.arduino.cc/hardware/nano-esp32/

[2] Analog Devices. How Common Noise and Error Sources Affect Optical Biosensing. Official technical note